COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 02

Combinational Circuit

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Comb. CKt.



H.A. , F.A.

Half Adder

- What is half adder and what are input lines and what are output lines ?

- I/P lines ⟶ 2- lines (x, y) ⟶ corresponding bits of the two nos being added.

$x \rightarrow a_3$, $x \rightarrow a_2$ $x \rightarrow a_1$

$y \rightarrow b_3$ $y \rightarrow b_2$ $y \rightarrow b_1$

$A = a_3\ a_2\ a_1\ a_0$

$B = b_3\ b_2\ b_1\ b_0$

$C_3\ C_2\ C_1$

$C_4\ S_3\ S_2\ S_1\ S_0$

O/P lines ⟶ Sum O/P S ⟶ After forwarding carry ('0' or '1') whatever is left that is sum o/p.

Carry O/P C ⟶ Will be '1' if addition of x & y will be greater than or equal to 2 (base) otherwise '0'.

	2^1	2^0		
	Input lines		Output lines	
	x	y	S-o/p	C-o/p
0	0	0	0	0
1	0	1	1	0
2	1	0	1	0
3	1	1	0	1

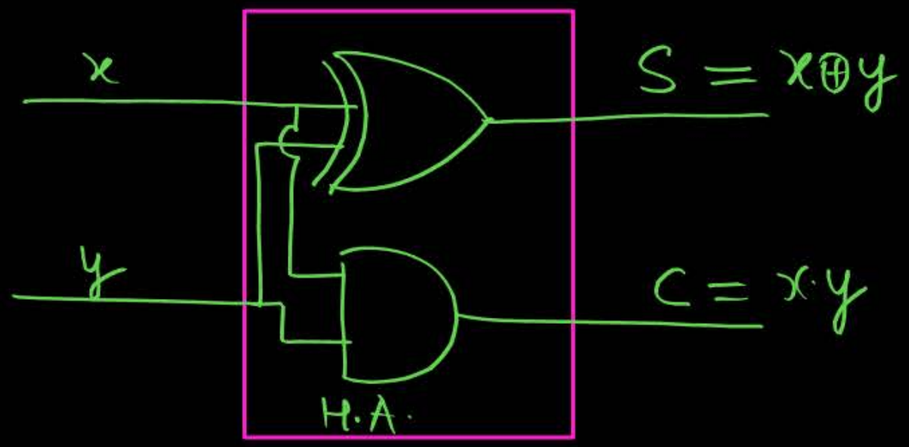
$S(x,y) = \Sigma(1,2) = \pi(0,3) = \bar{x}y + x\bar{y}$

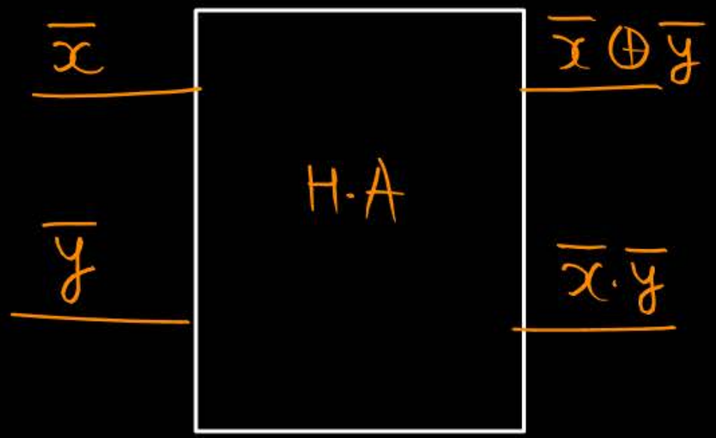
$= x \oplus y$

↳ non-self dual.

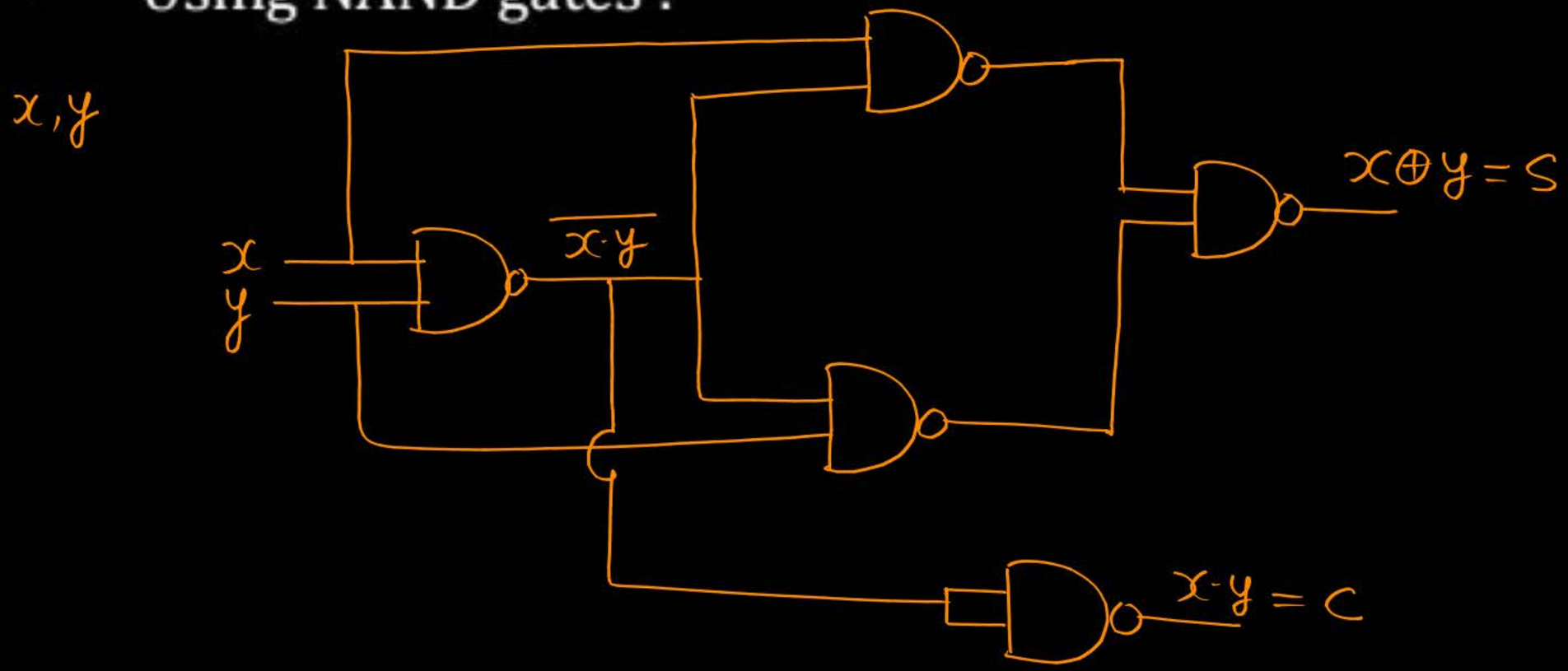
$C(x,y) = \Sigma(3) = x \cdot y$

↳ non-self dual.

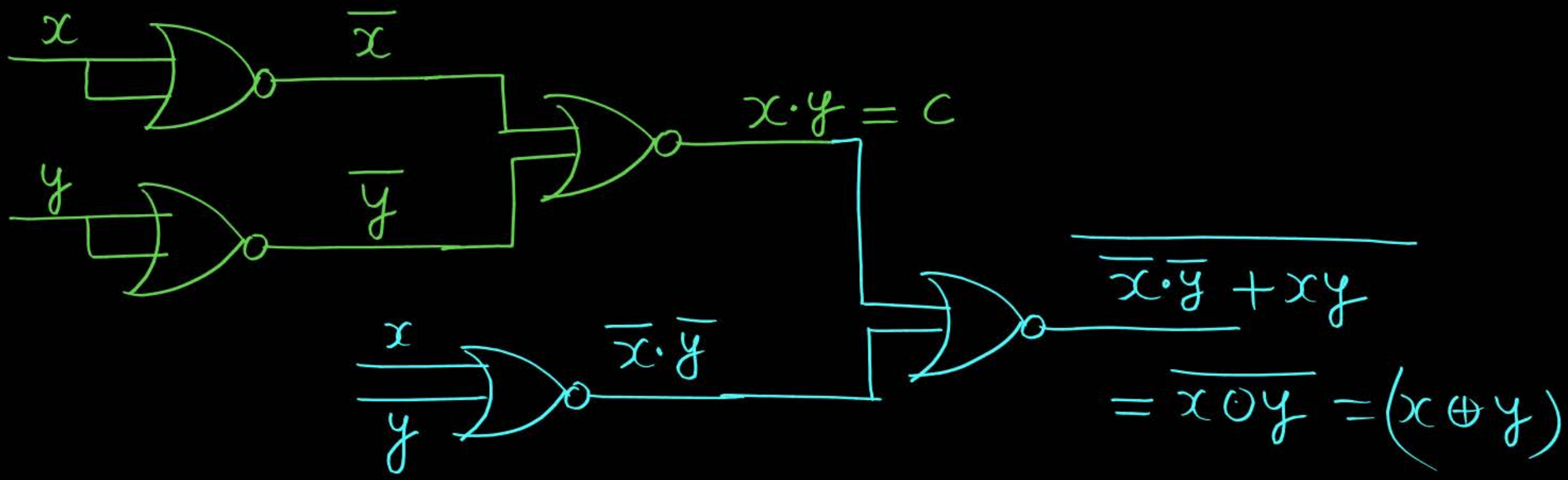




- Using NAND gates :



- Using NOR gates :



- To implement H.A we require either 5 (2-i/p) NAND gates or 5 (2-i/p) NOR gates.

- $A \cdot B \xleftrightarrow{\text{dual}} (A+B)$

- $A \oplus B \xleftrightarrow{\text{Dual}} A \odot B$

$= \bar{A}B + A\bar{B} \qquad (\bar{A}+B)\cdot(A+\bar{B})$

$A \odot B \xleftrightarrow{\text{Dual}} A \oplus B$

$\bar{A}\bar{B} + AB \qquad (\bar{A}+\bar{B})\cdot(A+B)$

$\overline{A \cdot B} \xleftrightarrow{\text{Dual}} \overline{A+B}$

NAND NOR

$f(x,y,z) = \Sigma(1,2,4,7)$

$= x \oplus y \oplus z$

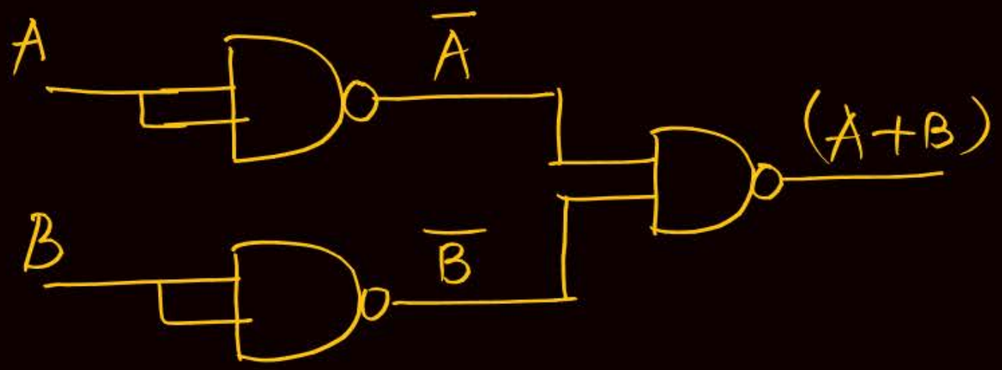
self dual

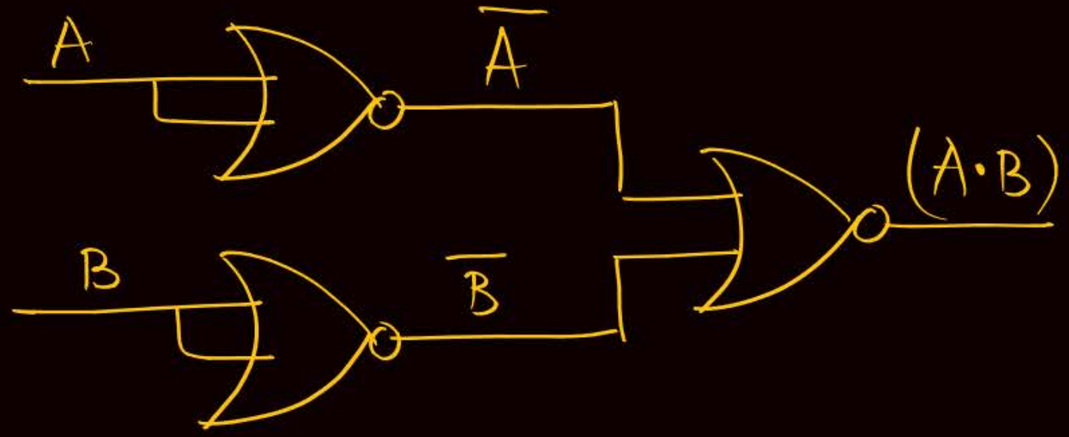
$x \oplus y \oplus z \xleftrightarrow{\text{Dual}} x \odot y \odot z$

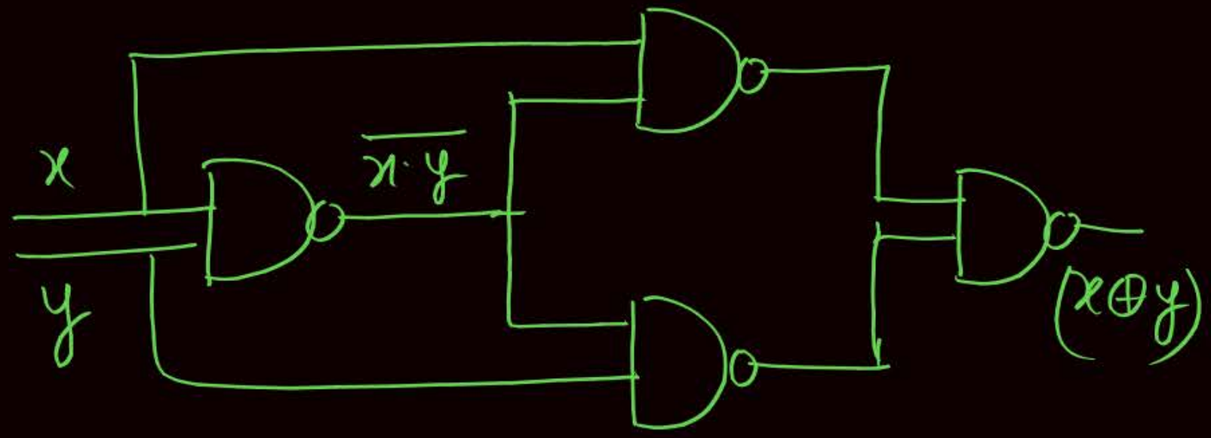
$A \oplus B \odot C \xrightarrow{\text{Dual}} A \odot B \oplus C$

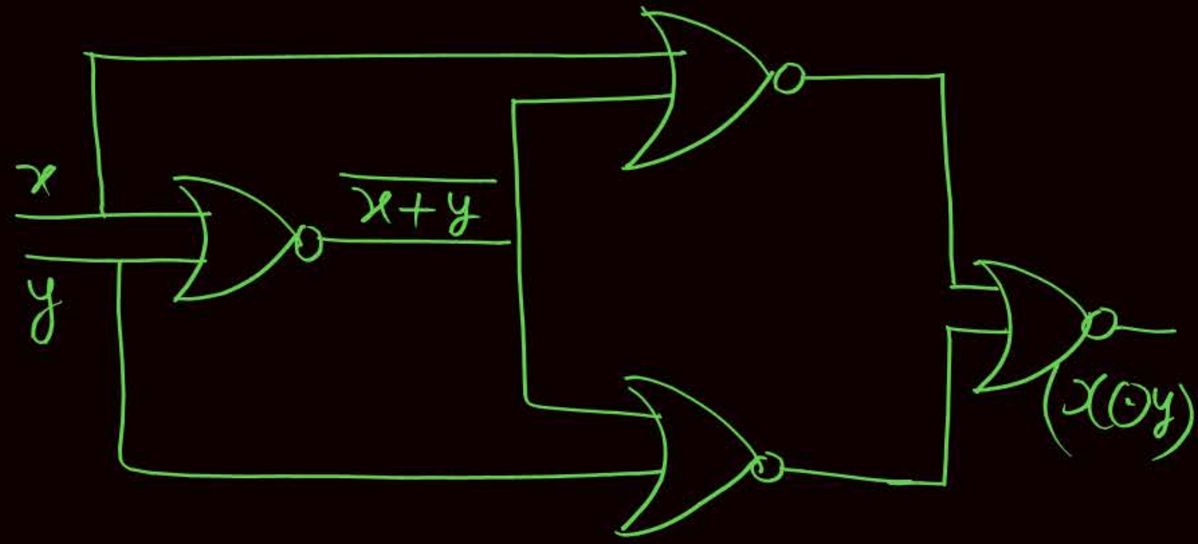
$\overline{A \oplus B \odot C} \qquad \overline{A \odot B \oplus C}$

$A \oplus B \odot C \rightarrow$ self dual.


A
$\bar{A}$
B
$\bar{B}$
$(\bar{A}+B)$


A
$\bar{A}$
B
$\bar{B}$
$(A \cdot B)$


x
y
$\overline{x \cdot y}$
$(x \oplus y)$


x
y
$\overline{x+y}$
$(x \odot y)$

Full Adder ***

- What is full adder and what are input lines and what are output lines ?

⟹ I/P lines ⟶ 3 - I/P lines ⟶ (x, y, z) ⟶ (x, y) corresponding bits of the two numbers being added

z ⟶ Carry O/P from the just previous addition.

O/P lines ⟶ 2 - O/P lines ⟶ Sum O/P S ⟶ After forwarding carry (0 or 1), whatever is left is sum O/P.

Carry O/P C ⟶ is '1' when addition x, y & z is greater than or equal to 2 (base) otherwise '0'.

	2^2	2^1	2^0		
	Input lines			Output lines	
	x	y	z	S-o/p	C-o/p
0	0	0	0	0	0
1	0	0	1	1	0
2	0	1	0	1	0
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	0	1
6	1	1	0	0	1
7	1	1	1	1	1

$C(x,y,z) = \bar{x}yz + x\bar{y}z + xy\bar{z} + xyz$

$= xy + z(x \oplus y)$

$S(x,y,z) = \Sigma(1,2,4,7) = x \oplus y \oplus z$

↳ self dual boolean function

$S^D = S$

***

$C(x,y,z) = \Sigma(3,5,6,7) = \pi(0,1,2,4)$

$C^D = C$

↳ self dual boolean function

majority I/P CKt

$= \bar{x}yz + x\bar{y}z + xy\bar{z} + xyz$

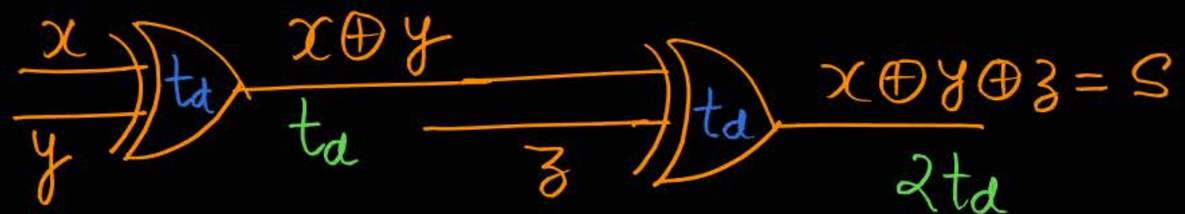
$= xy + xz + yz$

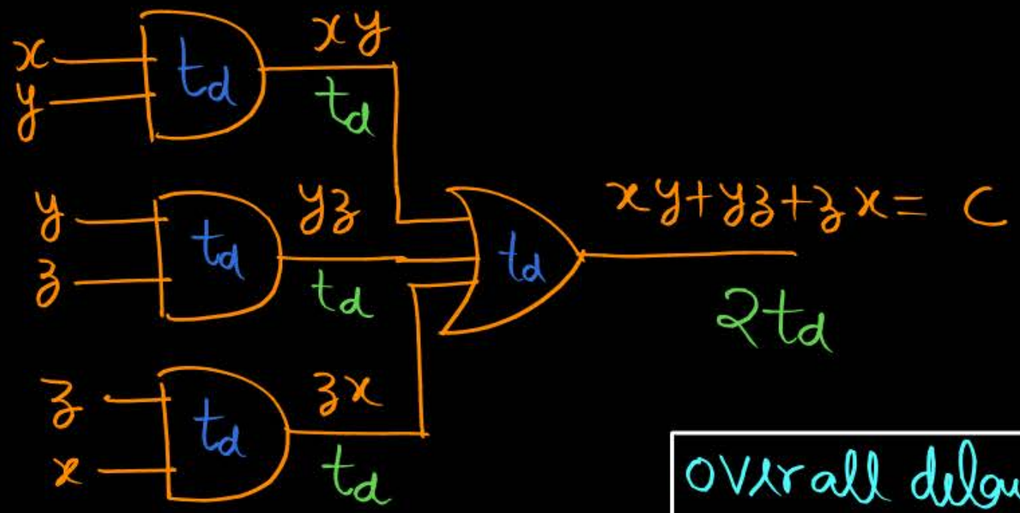
$= xy + yz + xz$

$= xy + z(x \oplus y)$

When majority i/p's are at logic '1' o/p is '1' and when majority i/p's are at '0' then o/p is '0'.

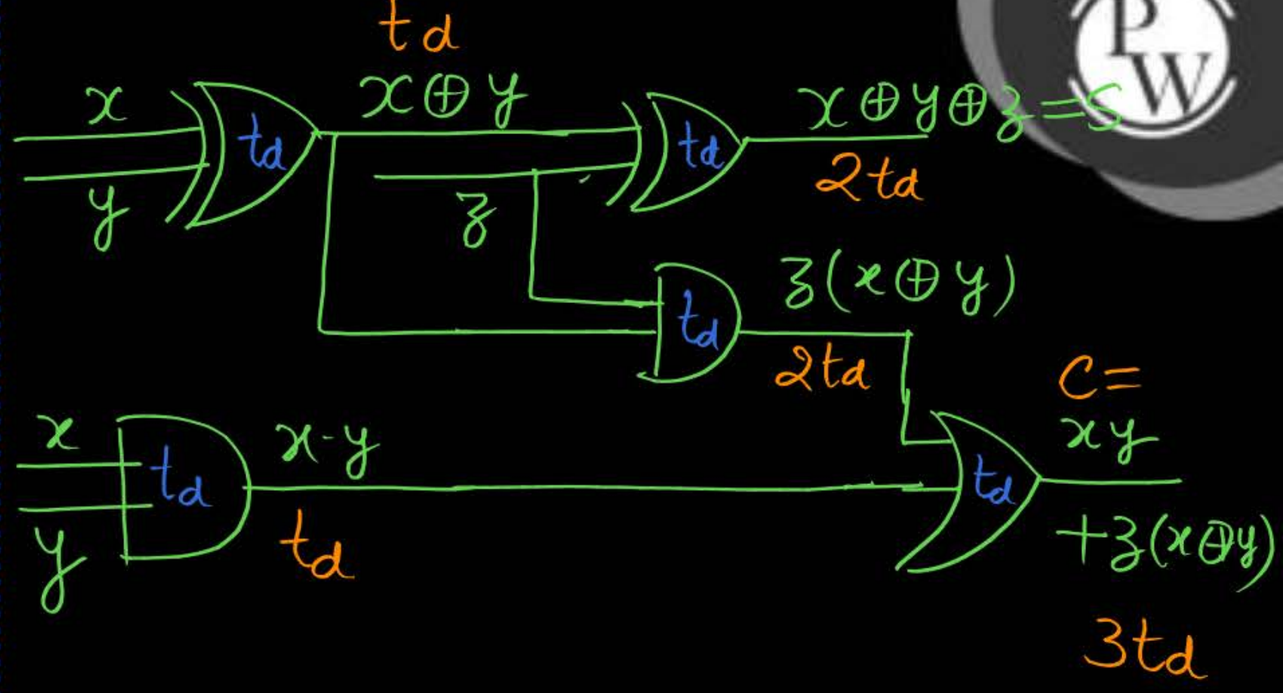
Implementation using gates :





CKt-I

Overall delay
$T_d = 2t_d$

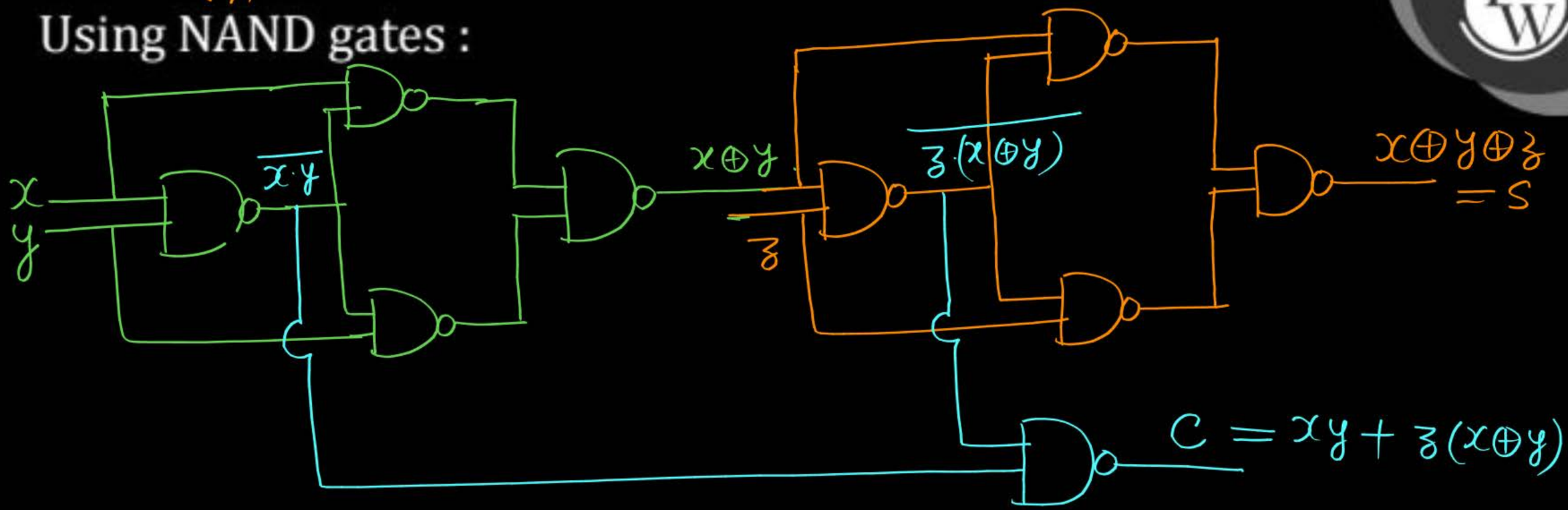


CKt - II

Overall delay $T_d = 3t_d$

2 i/p

- Using NAND gates :



- Using NOR gates :

H.W.

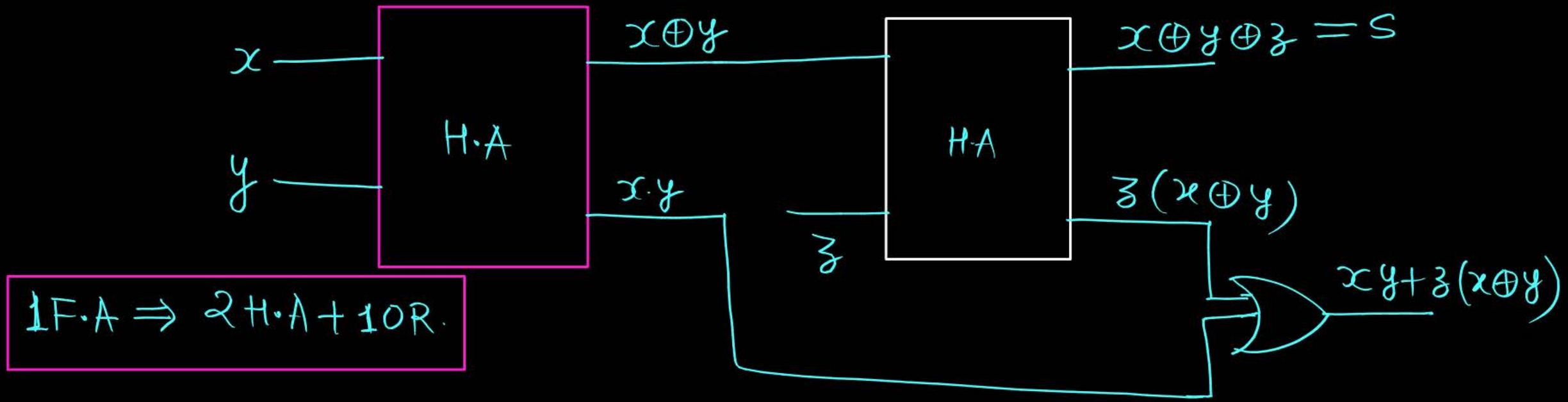
Note : To implement F.A. we require 9 [2-i/p] NAND or 9 [2-i/p] NOR gates.

- **Full adder using H.A. and OR gate :**

x, y, z → i/p lines

$$S = x \oplus y \oplus z$$

$$C = xy + yz + zx = xy + (x \oplus y)z$$

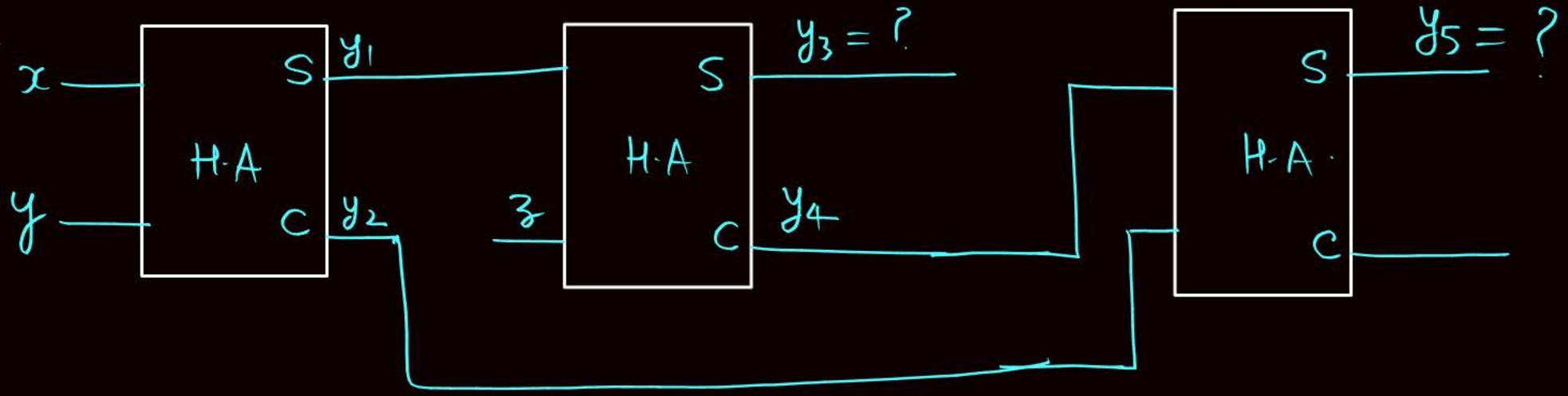


H.W

Q.1



Q.2





2 Minute Summary

- H.A
- F.A.

Thank you

GW

Soldiers !

